SQL Lab Questions

Dataset: CompanyDB

Table 1: Employees

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
102	Ravi	2	60000	2021-03-15	Mumbai
103	Neha	1	55000	2020-07-20	Delhi
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
106	Karan	3	72000	2018-08-25	Bangalore
107	Pooja	1	48000	2022-09-30	Delhi
108	Raj	2	52000	2021-12-11	Chennai
109	Meena	3	68000	2020-02-14	Pune
110	Vikas	1	51000	2023-01-18	Delhi

Table 2: Departments

dept_id	dept_name
1	HR
2	Sales
3	IT

Table 3: Projects

project_id	project_name	dept_id	budget
201	Alpha	1	100000
202	Beta	2	150000
203	Gamma	3	200000
204	Delta	2	120000

Table 4: Attendance

emp_id	date	status
101	2024-03-01	Present
102	2024-03-01	Absent
103	2024-03-01	Present
104	2024-03-01	Present
105	2024-03-01	Absent
106	2024-03-01	Present
107	2024-03-01	Present
108	2024-03-01	Present
109	2024-03-01	Absent
110	2024-03-01	Present

SQL Questions (Lab Practice)

Basic Level (1–10)

1. Retrieve all employees.
2. Display only emp_name and salary.
3. Find employees working in department 1.
4. List employees from Delhi.
5. Show employees with salary greater than 60000.
6. Find employees hired after 2021.
7. Display unique cities.
8. Sort employees by salary (descending).
9. Count total employees.
10. Find average salary.

Intermediate Level (11–20)

11. Find total salary paid per department.
12. Count employees in each department.
13. Find highest salary in each department.
14. Display employees along with department names.
15. List employees working on Sales department.
16. Find employees whose name starts with 'A'.
17. Get employees with salary between 50000 and 70000.
18. Find employees not from Mumbai.
19. Show employees hired in 2022.
20. Find top 3 highest paid employees.

Joins (21–25)

21. INNER JOIN Employees and Departments.
22. LEFT JOIN to show all employees with department names.
23. Find employees working in IT department.
24. Show project names with department names.
25. Find employees and their project budgets (via department).

Advanced Level (26–30)

26. Find second highest salary.
27. Rank employees by salary.
28. Find employees earning above average salary.
29. Get department with highest total salary.
30. Find employees who were absent on 2024-03-01.